Investigation: COMPANHIA ENERGÉTICA DE BRASÍLIA - CEB (CVM code 14451), fiscal year 2011, as it was on 2013-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Energia Elétrica (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2011 DFP, version 2, filed on 2012-04-09; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2012-03-31); 2 (2012-04-09).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=16053

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: component not computable: DEPI
- Alert by the study 1 rule: does not trigger alerts
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: —
- Percentile in group and year: —
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: low

DSRI, days' sales in receivables index
- Value: 1.012 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 45 (n = 52)
- Percentile in group and year: 45 (n = 457)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.028 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 71 (n = 53)
- Percentile in group and year: 59 (n = 467)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.008 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 55 (n = 52)
- Percentile in group and year: 53 (n = 462)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.073 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 41 (n = 53)

Percentile in group and year: 41 (n = 468)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: missing account: depreciation, cash flow lines in section 6.01.01
M-score component; it has no rule of its own
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low

SGAI, SG&A expenses index
Value: 1.050 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 62 (n = 51)
Percentile in group and year: 58 (n = 461)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.010 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 84 (n = 53)
Percentile in group and year: 64 (n = 484)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 0.977 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 25 (n = 53)
Percentile in group and year: 34 (n = 476)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 0.3 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 34.2 days)
Percentile in sector and year: 48 (n = 53)
Percentile in group and year: 45 (n = 441)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 9.26 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 81 (n = 51)
Percentile in group and year: 60 (n = 429)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.010 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 84 (n = 53)
Percentile in group and year: 65 (n = 484)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 0.355 (index)
Components: X1 = −0.092; X2 = −0.011; X3 = 0.066; X4 = 0.524 (indices)
Percentile in sector and year: 15 (n = 56)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 6/9 (8 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): 1
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 1
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 1
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Highlighted in the following year's comparative: 1 (threshold of R$21.7 million, 1% of total assets of the version used). Values in R$ million:
DFC (direct method) 6.01 Caixa Líquido Atividades Operacionais: 67.3 in the version used (v2, filed on 2012-04-09); 90.6 in the next DFP's comparative (v2, filed on 2013-03-28); difference of 23.3. Different value in the following year's comparative (restatement or reclassification).
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 35 accounts above the R$1,000 tolerance (group a, totals and key lines: 2; group b, reclassifications: 33), in the annex at the end.

What was known on the date

- FCA 2011, v1: 2011-01-10
- FCA 2011, v2: 2011-04-01
- FCA 2011, v3: 2011-05-09
- DFP 2010, v1: 2011-07-15
- FCA 2011, v4: 2011-07-15
- DFP 2010, v2: 2011-07-18
- FCA 2011, v5: 2011-09-01
- FCA 2011, v6: 2011-10-18
- DFP 2011, v1: 2012-03-31
- DFP 2011, v2: 2012-04-09
- DFP 2012, v1: 2013-03-27
- DFP 2012, v2: 2013-03-28

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2010 FCA: KPMG Auditores Independentes.
Auditor in the 2011 FCA: KPMG Auditores Independentes.
Auditor in the 2012 FCA: KPMG Auditores Independentes.
Auditor changes: none in the FCAs filed by the date.
Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and

location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:31:45+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

Annex: full list of differences in the following year's comparative (R$ million)

(a) DFC (direct method) 6.01 Caixa Líquido Atividades Operacionais: 67.3 in the version used, 90.6 in the comparative (difference of 23.3)
(a) DRE 3.01 Receita de Venda de Bens e/ou Serviços: 1,377.6 in the version used, 1,375.4 in the comparative (difference of −2.2)
(b) BPA 1.02.04.01 Intangíveis: 388.4 in the version used, 0.0 in the comparative (difference of −388.4)
(b) BPA 1.02.04.01.01 Contrato de Concessão: 249.9 in the version used, 0.0 in the comparative (difference of −249.9)
(b) BPA 1.02.03.01 Imobilizado em Operação: 193.1 in the version used, 0.0 in the comparative (difference of −193.1)
(b) BPA 1.02.04.01.03 Goodwill: 123.6 in the version used, 0.0 in the comparative (difference of −123.6)
(b) BPP 2.01.01 Obrigações Sociais e Trabalhistas: 75.8 in the version used, 22.7 in the comparative (difference of −53.1)
(b) BPP 2.01.01.02 Obrigações Trabalhistas: 75.8 in the version used, 22.7 in the comparative (difference of −53.1)
(b) BPP 2.01.01.02.03 Benefício a Empregados: 53.1 in the version used, 0.0 in the comparative (difference of −53.1)
(b) BPP 2.01.05 Outras Obrigações: 165.9 in the version used, 218.9 in the comparative (difference of 53.1)
(b) BPP 2.01.05.02 Outros: 165.9 in the version used, 218.9 in the comparative (difference of 53.1)
(b) BPP 2.01.05.02.06 Consumidores: 0.0 in the version used, 53.1 in the comparative (difference of 53.1)
(b) BPA 1.01.03 Contas a Receber: 357.2 in the version used, 306.2 in the comparative (difference of −51.0)
(b) BPA 1.01.03.01 Clientes: 357.2 in the version used, 306.2 in the comparative (difference of −51.0)
(b) BPA 1.01.03.01.09 Outros Créditos: 51.0 in the version used, 0.0 in the comparative (difference of −51.0)
(b) BPA 1.01.08 Outros Ativos Circulantes: 0.0 in the version used, 51.0 in the comparative (difference of 51.0)
(b) BPA 1.01.08.03 Outros: 0.0 in the version used, 51.0 in the comparative (difference of 51.0)
(b) DFC (direct method) 6.01.02 Fornecedores - Materiais e Serviços: −198.0 in the version used, −174.7 in the comparative (difference of 23.3)
(b) DFC (direct method) 6.02 Caixa Líquido Atividades de Investimento: −37.4 in the version used, −60.7 in the comparative (difference of −23.3)
(b) DFC (direct method) 6.02.01 Aquisição de Imobilizado: −57.9 in the version used, −81.1 in the comparative (difference of −23.3)
(b) BPP 2.01.01.02.01 Folha de Pagamento: 22.7 in the version used, 0.0 in the comparative (difference of −22.7)
(b) BPA 1.02.01.09 Outros Ativos Não Circulantes: 653.9 in the version used, 676.4 in the comparative (difference of 22.5)
(b) BPA 1.02.01.03 Contas a Receber: 134.5 in the version used, 117.6 in the comparative

(difference of −17.0)
(b) BPA 1.02.01.03.02 Outras Contas a Receber: 17.0 in the version used, 0.0 in the comparative (difference of −17.0)
(b) BPA 1.02.04.01.04 Outros Intangíveis: 14.8 in the version used, 0.0 in the comparative (difference of −14.8)
(b) BPP 2.01.06.01.06 Outras Provisões: 11.5 in the version used, 0.0 in the comparative (difference of −11.5)
(b) BPA 1.02.01.01 Aplicações Financeiras Avaliadas a Valor Justo: 15.9 in the version used, 10.4 in the comparative (difference of −5.5)
(b) BPA 1.02.01.01.04 Depósitos Vinculados a Litígios: 5.5 in the version used, 0.0 in the comparative (difference of −5.5)
(b) DRE 3.04.05 Outras Despesas Operacionais: −76.2 in the version used, −81.1 in the comparative (difference of −4.9)
(b) DRE 3.05 Resultado Antes do Resultado Financeiro e dos Tributos: 144.1 in the version used, 139.3 in the comparative (difference of −4.9)
(b) DRE 3.06 Resultado Financeiro: −63.6 in the version used, −58.7 in the comparative (difference of 4.9)
(b) DRE 3.06.02 Despesas Financeiras: −95.5 in the version used, −90.6 in the comparative (difference of 4.9)
(b) DRE 3.04 Despesas/Receitas Operacionais: −163.8 in the version used, −166.4 in the comparative (difference of −2.7)
(b) DRE 3.03 Resultado Bruto: 307.9 in the version used, 305.7 in the comparative (difference of −2.2)
(b) DRE 3.04.04 Outras Receitas Operacionais: 115.3 in the version used, 117.5 in the comparative (difference of 2.2)